



Linux File Permissions – Deep Dive

This document provides an in-depth explanation of Linux file permissions. File permissions are a core security mechanism in Linux and are critical for DevOps engineers, Cloud administrators, and Cybersecurity professionals.

1. Understanding Linux File Permissions

Linux controls access to files and directories using a permission model. Every file and directory has defined permissions that specify who can read, write, or execute it.

Permissions apply to:

- File owners
- Groups
- Others (everyone else)

2. Permission Types

Read (r) – View file content or list directory contents

Write (w) – Modify file or directory

Execute (x) – Run file or access directory

3. Viewing Permissions

```
ls -l
```

Displays file permissions, ownership, and metadata.

Example output:

```
-rwxr-xr--
```

4. Permission Structure Explained

Example: -rwxr-xr--

- First character: file type
- rwx – owner permissions
- r-x – group permissions
- r-- – others

5. chmod – Changing Permissions

```
chmod 755 script.sh
```

Sets read, write, execute for owner; read and execute for group and others.

```
chmod u+x file.sh
```

Symbolic mode: adds execute permission for user.

6. chown & chgrp – Ownership Control

```
chown user file.txt
```

Changes file owner.

```
chgrp group file.txt
```

Changes group ownership.

7. Special Permissions

SUID – Execute as file owner

SGID – Execute as group owner

Sticky Bit – Restrict deletion in shared directories

`chmod 4755 file`

8. Permissions in DevOps, Cloud & Cybersecurity

DevOps: Prevent pipeline failures due to permission errors.

Cloud: Secure EC2 files, SSH keys, and configs.

Cybersecurity: Reduce attack surface and prevent privilege escalation.